

What Are You Proving?

Week 1 Assignment — Interview Transcript (Claude as thinking partner)

Opening frame

Q1. What can you genuinely build today?

Described RepoMindAI: a RAG-based developer intelligence system over GitHub repositories. After core RAG functionality worked, containerized with Docker, added Grafana-based monitoring/observability, and built a CI/CD workflow running Ruff linting and Pytest. Explicitly noted no cloud deployment yet — a boundary drawn honestly rather than blurred.

Q2. What kind of AI engineering work do you actually want to be hired for?

Most interested in building RAG/LLM-powered applications that solve practical problems, not primarily training or fine-tuning models from scratch. RepoMind reflects this: the core problem is making GitHub repository knowledge searchable/usable via RAG, plus the surrounding software system (APIs, Docker, testing, CI/CD, observability). Enjoys the engineering side of making AI systems usable and maintainable, not just the model itself.

Q3. What differentiates you from another student who says they know AI/ML?

The real technical problem: RepoMind ingests heterogeneous repo content (.py, .md, config files, docs, eventually issues/PRs/setup info) while trying to preserve structure and context. For retrieval, actively comparing strategies (e.g., document-level chunking, preserving repo/file metadata) rather than assuming one approach is best, and evaluating retrieval quality using Hit Rate and MRR — not just eyeballing whether generated answers look good. Generation-level evaluation not yet done. Framed the differentiator as: not building “a RAG chatbot over GitHub,” but figuring out how to represent and retrieve knowledge from an actual software repository with enough structure for results to be useful, and testing whether the retrieval strategy actually works.

Q4. What evidence do you actually have?

Retrieval evaluation is reviewable today: a sample/generated dataset with current results documented in the README, using Hit Rate and MRR to compare retrieval strategies. Explicitly still V1 — generation-level evaluation and live-user monitoring are not done yet. Clear line drawn between what's checkable right now (retrieval eval) versus what's still in progress (broader evaluation, live observability).

Q5. Who specifically do you want the portfolio to convince?

A hiring manager at a small-to-mid-sized startup actually building AI-powered products — for an internship where the expectation is building things and contributing to the surrounding software, not primarily ML research or training models from scratch. This matches what RepoMind demonstrates:

RAG/LLM systems, retrieval, evaluation, backend engineering, and the infrastructure that makes those systems usable.

Q6. What single action do you want that person to take?

Kept the action centered on reviewing project work broadly rather than pointing only at RepoMindAI, because the portfolio also includes AWS projects (S3, CloudFront, Lambda, Polly) and a Kubernetes-based e-commerce microservices project that demonstrate the cloud/infrastructure capability needed to take AI systems further. Wanted project case studies to be the main path through the site, not a single project alone.

Pushback: does that dilute the AI claim?

Challenged whether treating all projects as equal weight (via a flat “explore my projects” action) undercuts the specific AI-engineering claim, since the AWS/Kubernetes projects contain no AI work and prove a broader, different claim (cloud/DevOps capability). Asked directly: is “explore my projects” because all projects genuinely deserve equal weight, or because of uncertainty that RepoMindAI is “enough” on its own?

Resolution

Agreed RepoMindAI should be the primary proof of the AI engineering claim, with AWS and Kubernetes projects positioned as supporting evidence of the engineering foundation underneath that direction (not competing proof, and not claimed as AI engineering themselves). Portfolio hierarchy: primary proof (RepoMindAI) versus supporting evidence (cloud/infrastructure projects).

Q7. What can this portfolio prove that your CV/LinkedIn cannot?

A CV bullet on RepoMindAI would name the RAG system, Docker, CI/CD, and retrieval evaluation, but cannot show the reasoning: how heterogeneous repo content was handled while preserving structure, why different retrieval/chunking strategies were compared, how retrieval quality was actually evaluated with Hit Rate and MRR, or the engineering trade-offs and limitations encountered while building V1. A portfolio makes that reasoning inspectable through architecture, case study, evaluation results, and implementation detail — instead of asking a hiring manager to take resume claims at face value.

Q8. Is “productionize” the right verb?

Weighed “productionize” against the actual evidence: Docker, CI/CD, early-stage Grafana observability, and retrieval evaluation exist, but cloud deployment, background ingestion, and generation-level evaluation do not. Concluded “productionize” overclaims. Considered and rejected “build and engineer” (too generic, loses the applied-AI focus) and “build and evaluate” (drops the software engineering side). Landed on: “I build and engineer applied AI systems” as the most honest description of what can be proven today, with the explicit intent to strengthen the claim later as cloud deployment, better observability, and generation-level evaluation are added.

Working Summary (draft direction — final wording left to the student)

Primary claim:

You build and engineer applied AI systems — specifically RAG/LLM-powered applications — with real software engineering practices around them (containerization, CI/CD, retrieval evaluation, early observability), rather than stopping at a working notebook or claiming full cloud production deployment not yet done.

Specific person:

A hiring manager at a small-to-mid-sized startup building AI-powered products, hiring for an AI engineering internship focused on building applied systems (not primarily ML research or training models from scratch).

Primary action:

Review the project work, starting with RepoMindAI as the primary proof of the AI claim, with cloud/infrastructure projects (AWS, Kubernetes) visible as supporting evidence of the engineering foundation underneath it — not competing with it.

Why this portfolio needs to exist:

A CV can list “RAG system, Docker, CI/CD, retrieval evaluation” as bullet points, but it cannot show the reasoning — why chunking/retrieval strategies were compared, how retrieval quality was evaluated with Hit Rate/MRR against a documented dataset, or the trade-offs and limitations of a V1 system. The portfolio makes that reasoning and evidence inspectable instead of asking the hiring manager to take resume claims on faith.

Open items flagged for the final write-up: keep the RAG/retrieval-evaluation specificity from Q3 so “applied AI systems” doesn’t flatten back into something generic; state RepoMindAI’s V1 stage plainly rather than letting a hiring manager discover the gaps on their own; and make sure the action sentence implies the RepoMindAI-first hierarchy rather than reading as a flat, undifferentiated project list.